

UNIVERSIDADE ESTADUAL DE MONTES CLAROS – UNIMONTES
CENTRO DE CIÊNCIAS EXATAS E TECNOLÓGICAS – CCET
DEPARTAMENTO DE CIÊNCIAS DA COMPUTAÇÃO – DCC

ATIVIDADE AVALIATIVA II

RAMON LOPES DE QUEIROZ

MONTES CLAROS – MG

ABRIL/2026

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ATIVIDADE AVALIATIVA II

Atividade avaliativa apresentada para atendimento de requisito parcial para aprovação na disciplina Geometria Analítica e Álgebra Linear do Curso de Graduação em Bacharelado em Sistemas de Informação – 2º período

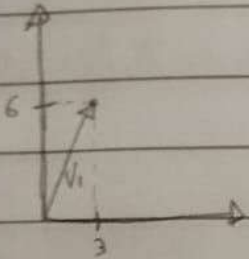
Professor: Antônio Wilson Vieira

MONTES CLAROS – MG

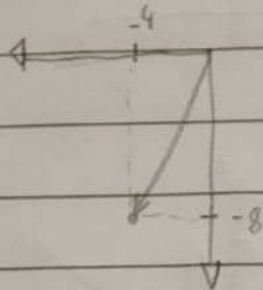
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Exercício - Decisão 3.1 - 3, 5, 7, 9, 13, 26, 31 -

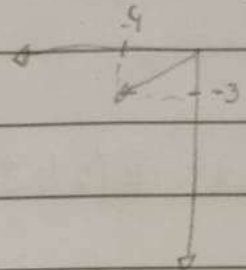
3-a. $V_1 = (3, 6)$



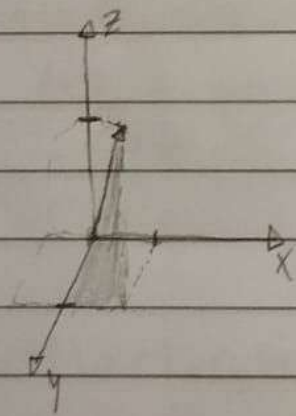
b. $V_2 = (-4, -8)$



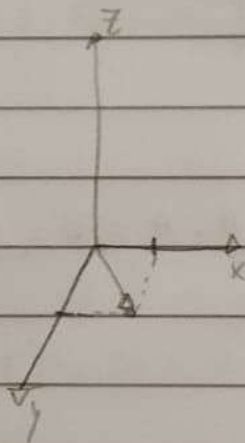
c. $V_3 = (-4, -3)$



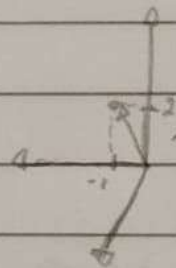
d. $V_4 = (3, 4, 5)$



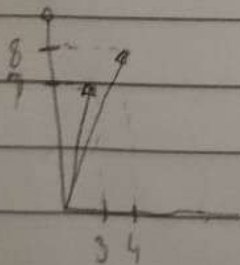
e. $V_5 = (3, 3, 0)$



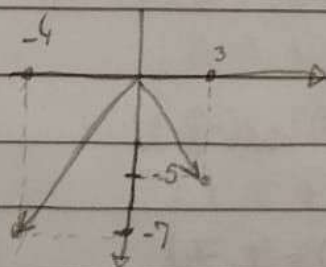
f. $V_6 = (-1, 0, 2)$



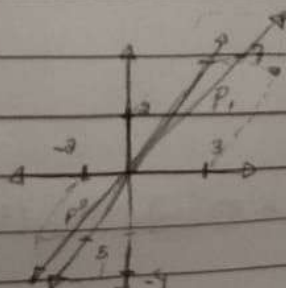
5-a. $P_1(4, 8), P_2(3, 7)$



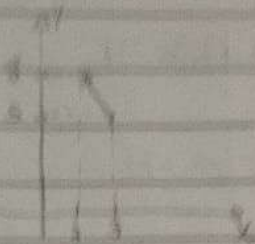
b. $P_1(3, -5), P_2(-4, -7)$



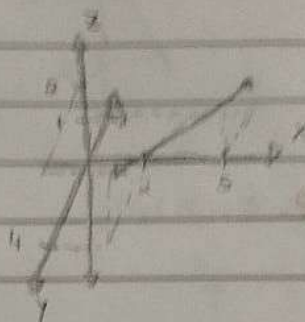
c. $P_1(3, -7, 2), P_2(-2, 5, -4)$



7-a. $P_1(1,5), P_2(2,2)$



8. $P_1(5,-2,1), P_2(2,4,2)$



9a. $A(1,1), U(1,2)$

$$u = B - A \Rightarrow B = u + A$$

$$V = (1+1, 1+2) \Rightarrow V = (2, 3)$$

10. $B(-1,-1,2), U(1,1,3)$

$$A = B - U$$

$$V = (-1-1, -1-1, 2-3) \Rightarrow V = (-2, -2, -1)$$

13-a. $u = (4,-1), v = (0,5), w = (-3,-3)$

$$u + w = (4,-1) + (-3,-3) = (1,-4)$$

b. $(0,5) - 3 \cdot (4,-1)$
 $(-12, 8)$

c. $2u - 10w = 2(4,-1) - 10(-3,-3) = (8,-2) - (-30,-30) = (38, 28)$

$$d. -3v - 2(u + 2w) = (4, -1) + 2(-3, -3) + 3v \\ 3(0, 5) - 2(-2, -7) \\ (4, 29)$$

$$e. -3(w - 2u + v) = 3((-3, -3) - 2(4, -1) + (0, 5)) \\ -3(-11, 4) \\ (33, -12)$$

$$f. (-2u - w) + 5(v + 3w) \\ -2(4, -1) - (0, 5) + 5((0, 5) + 3(-3, -3)) \\ (-8, -3) + (45, -20) \\ (37, 17)$$

$$26. c_1(1, 2, 0) + c_2(2, 1, 1) + c_3(0, 3, 1) = (0, 0, 0)$$

$$c_1 = 0$$

$$c_2 = 0$$

$$c_3 = 0$$

Three linearly independent.

$$\begin{cases} a + 2b + 0 = 0 & 1 \ 2 \ 0 \ 0 \\ 2a + b + 3c = 0 & 2 \ 1 \ 3 \ 0 \\ 0 + b + c = 0 & 0 \ 1 \ 1 \ 0 \end{cases}$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad b = -c \\ a + 2(-c) = 0, \quad 2c$$

$$2(-c) - c + 3c = 0 \\ c = 0$$

1 1 1

31- $C_1(-2, 9, 6) + C_2(-3, 2, 1) + C_3(1, 7, 5) = (0, 5, 4)$

$$\begin{cases} -2a - 3b + c = 0 \\ 9a + 2b + 7c = 5 \\ 6a + b + 5c = 4 \end{cases} \Rightarrow \begin{bmatrix} -2 & -3 & 1 & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow{\substack{-\frac{1}{2}L_1 \\ \frac{1}{2}L_1}} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 9 & 2 & 7 & 5 \\ 6 & 1 & 5 & 4 \end{bmatrix} \xrightarrow{\substack{L_2 - 9L_1 \\ L_3 - 6L_1}} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & -\frac{23}{2} & \frac{23}{2} & 5 \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{\substack{-\frac{2}{23}L_2 \\ \frac{2}{23}L_2}} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & -8 & 8 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & -8 & 8 & 4 \end{bmatrix} \xrightarrow{L_3 + 8L_2} \begin{bmatrix} 1 & \frac{3}{2} & -\frac{1}{2} & 0 \\ 0 & 1 & -1 & -\frac{10}{23} \\ 0 & 0 & 0 & \frac{12}{23} \end{bmatrix}$$

SI $\rightarrow$ Não tem escalonamento possível.

Exercício - Resolução 3.2 (Ímpares) -

1-a. $V = (4, -3)$ $\|V\| = \sqrt{4^2 + (-3)^2} = \sqrt{25} = 5$

$$u = \frac{1}{\|V\|} \cdot V = \frac{1}{5} \cdot (4, -3) = \left(\frac{4}{5}, -\frac{3}{5}\right)$$

$$-u = \left(-\frac{4}{5}, \frac{3}{5}\right) = \left(\frac{-4}{5}, \frac{3}{5}\right)$$

b. $V = (2, 2, 2)$ $\|V\| = \sqrt{2^2 + 2^2 + 2^2} = \sqrt{12} \approx 3,46 = 2\sqrt{3}$

$$u = \frac{1}{2\sqrt{3}} (2, 2, 2) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$-u = \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}\right)$$

c. $V = (1, 0, 2, 1, 3)$ $\|V\| = \sqrt{1^2 + 0^2 + 2^2 + 1^2 + 3^2} = \sqrt{15}$

$$u = \left(\frac{1}{\sqrt{15}}, 0, \frac{2}{\sqrt{15}}, \frac{1}{\sqrt{15}}, \frac{3}{\sqrt{15}}\right)$$

$$-u = \left(-\frac{1}{\sqrt{15}}, 0, -\frac{2}{\sqrt{15}}, -\frac{1}{\sqrt{15}}, -\frac{3}{\sqrt{15}}\right)$$

3.a. $u = (2, -2, 3)$ $v = (1, -3, 4)$ $w = (3, 6, -4)$

$$u+v = (2, -2, 3) + (1, -3, 4) = (3, -5, 7)$$

$$\|u+v\| = \sqrt{3^2 + (-5)^2 + 7^2} = \sqrt{83}$$

d. $\|u\| + \|v\|$
 $\sqrt{2^2 + (-2)^2 + 3^2} + \sqrt{1^2 + (-3)^2 + 4^2}$
 $\sqrt{17} + \sqrt{26}$

c. $\| -2u + 2v \| = 2 \| -u + v \|$
 $2 \cdot \sqrt{(-1)^2 + (-1)^2 + 1^2}$
 $2\sqrt{3}$

$$d \quad \|3u - 5v + w\|$$

$$\|(4, 15, -15)\| = \sqrt{4^2 + 15^2 + (-15)^2} = \sqrt{466}$$

$$3 \cdot (2, -2, 3) - 5(1, -3, 4) + (3, 6, -4)$$

$$(6, -6, 9) - (5, -15, 20) + (3, 6, -4)$$

$$(4, 15, -15)$$

$$5-a. \quad u = (-2, -1, 4, 5) \quad v = (3, 1, -5, 7) \quad w = (-6, 2, 1, 1)$$

$$3 \cdot (-2, -1, 4, 5) - 5 \cdot (3, 1, -5, 7) + (-6, 2, 1, 1)$$

$$(-6, -3, 12, 15) - (15, 5, -25, 35) + (-6, 2, 1, 1)$$

$$(-27, -6, 38, -19)$$

$$\|(-27, -6, 38, -19)\| = \sqrt{(-27)^2 + (-6)^2 + (38)^2 + (-19)^2} = \sqrt{729 + 36 + 1444 + 361} = \sqrt{2570}$$

$$b. \quad \|3u\| - 5\|v\| + \|w\|$$

$$\|(-6, -3, 4, 5)\| - 5\|(3, 1, -5, 7)\| + \|(-6, 2, 1, 1)\|$$

$$\sqrt{86} - 5\sqrt{94} + \sqrt{42}$$

$$c. \quad \| -\|u\| \cdot v \|$$

$$\| -\|(-2, -1, 4, 5)\| \cdot (3, 1, -5, 7) \|$$

$$\| -(\sqrt{46}) \cdot (3, 1, -5, 7) \|$$

$$\| -3\sqrt{46}, -\sqrt{46}, 5\sqrt{46}, -7\sqrt{46} \|$$

$$\sqrt{(-3\sqrt{46})^2 + (-\sqrt{46})^2 + (5\sqrt{46})^2 + (-7\sqrt{46})^2}$$

$$\sqrt{414 + 46 + 1150 + 2254}$$

$$\sqrt{3864}$$

$$7. v = (-2, 3, 0, 6)$$

$$\|v\| = \sqrt{(-2)^2 + 3^2 + 0^2 + 6^2} = \sqrt{49} = 7$$

$$\|Kv\| = 5$$

$$K \cdot \|v\| = 5$$

$$K \cdot 7 = 5$$

$$K = \pm \frac{5}{7}$$

$$9.a. (3, 1, 4) \cdot (2, 2, -4)$$

$$(3 \cdot 2, 1 \cdot 2, 4 \cdot -4)$$

$$(6, 2, -16)$$

$$6 + 2 - 16$$

$$-8$$

$P_6 =$

produto

$$(3, 1, 4) \cdot (3, 1, 4)$$

$$(9, 1, 16)$$

$$9 + 1 + 16$$

$$26$$

$$(2, 2, -4) \cdot (2, 2, -4)$$

$$(4, 4, 16)$$

$$24$$

$$1e. (1, 1, 4, 6) \cdot (2, 2, 3, 2)$$

$$2 - 2 + 12 - 12$$

$$0$$

$$(1, 1, 4, 6) \cdot (1, 1, 4, 6)$$

$$1, 1, 16, 36$$

$$54$$

$$(2, 2, 3, 2) \cdot (2, 2, 3, 2)$$

$$4 + 4 + 9 + 4$$

$$21$$

$$11a. u = (3, 3, 3), v = (1, 0, 4)$$

$$\sqrt{(3-1)^2 + (3-0)^2 + (3-4)^2}$$

$$\sqrt{2^2 + 3^2 + (-1)^2}$$

$$\sqrt{14}$$

$$b. u = (0, -2, -1, 1), v = (-3, 2, 4, 4)$$

$$\sqrt{(0+3)^2 + (-2-2)^2 + (-1-4)^2 + (1-4)^2}$$

$$\sqrt{9 + 16 + 25 + 9}$$

$$\sqrt{59}$$

$$c. u = (3, -3, -2, 0, 3, 13, 5), v = (-4, 1, -1, 5, 0, -11, 4)$$

$$\sqrt{(3+4)^2 + (-3-1)^2 + (-2+1)^2 + (0-5)^2 + (3-0)^2 + (13+11)^2 + (5-4)^2}$$

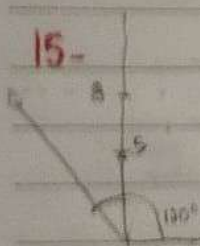
$$49 + 16 + 1 + 25 + 9 + 576 + 1$$

$$\sqrt{677}$$

13. a. $\cos \theta = \frac{U \cdot V}{\|U\| \|V\|} = \frac{(3, 1) \cdot (3, 0) + (3, 4)}{\sqrt{9+1} \sqrt{9+0+6}} = \frac{15}{\sqrt{10} \sqrt{15}} = \frac{15}{3\sqrt{3} \cdot \sqrt{5}} = \frac{15}{3\sqrt{15}} = \frac{5}{\sqrt{15}}$ Positivo = ângulo agudo

b. $\cos \theta = \frac{(0, -3) + (-2, 2) + (-1, 4) + (1, 4)}{\sqrt{0+4+1+1} \cdot \sqrt{9+4+16+16}} = \frac{-4}{\sqrt{6} \cdot \sqrt{45}}$ Negativo = Obtuso

c. $\cos \theta = \frac{(3, -4) + (-2, 1) + (-2, -1) + (0, 9) + (-3, 0) + (13, -11) + (5, 4)}{\sqrt{9+9+4+0+9+16+25} \cdot \sqrt{16+4+1+25+0+121+16}} = \frac{-136}{\sqrt{725} \cdot \sqrt{170}}$ Negativo = Obtuso



$|a| = 9, 120^\circ$

$120 - 90 = 30^\circ$

$|b| = 5, 90^\circ$

$9 \cdot 5 \cdot \cos(30^\circ)$

$\frac{45 \cdot \sqrt{3}}{2}$

$a \cdot b = 22,5\sqrt{3} \approx 38,97$

17. a. $U \cdot (V \cdot W) = \text{Correto}$. $V \cdot W$ resulta em um escalar, e a multiplicação escalar se aplica entre dois vetores, e não entre escalar e vetor.

b. $U \cdot (V + W) = \text{Correto}$.

c. $\|U \cdot V\| \neq \text{Não}$. Pois a norma é pra vetores, e a multiplicação resulta em um escalar.

d. $(U \cdot V) \cdot W = \text{Correto}$.

$$19. a. (-4, -3) = \sqrt{(-4)^2 + (-3)^2} = \sqrt{25} = 5 \quad u = \left(\frac{-4}{5}, \frac{-3}{5} \right)$$

$$b. (1, 7) = \sqrt{1 + 49} = \sqrt{50} \quad u = \left(\frac{1}{\sqrt{50}}, \frac{7}{\sqrt{50}} \right)$$

$$c. (-3, 2, \sqrt{3}) = \sqrt{(-3)^2 + 2^2 + (\sqrt{3})^2} = \sqrt{16} = 4 \quad u = \left(\frac{-3}{4}, \frac{2}{4}, \frac{\sqrt{3}}{4} \right)$$

$$d. (1, 2, 3, 4, 5) = \sqrt{1^2 + 2^2 + 3^2 + 4^2 + 5^2} = \sqrt{55} \quad u = \left(\frac{1}{\sqrt{55}}, \frac{2}{\sqrt{55}}, \frac{3}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{5}{\sqrt{55}} \right)$$

$$21. u = \frac{v}{\|v\|} \quad w = m \cdot u$$

$$w = m \cdot \frac{v}{\|v\|}$$

$$23. a. u = (2, 3), v = (5, -7)$$

$$\frac{2 \cdot 5 + 3 \cdot (-7)}{\sqrt{2^2 + 3^2} \cdot \sqrt{5^2 + (-7)^2}} = \frac{-11}{\sqrt{13} \cdot \sqrt{74}} = \frac{-11}{\sqrt{962}}$$

$$b. \cos \theta = \frac{-6 \cdot 4 + (-2) \cdot 0}{\sqrt{36 + 4} \cdot \sqrt{16 + 0}} = \frac{-24}{\sqrt{40} \cdot \sqrt{16}} = \frac{-24}{4\sqrt{40}}$$

$$c. \cos \theta = \frac{1 \cdot 3 + 5 \cdot 3 + 4 \cdot 3}{\|u\| \cdot \|v\|} = 0 \quad \therefore \cos \theta = 0$$

$$d. \cos \theta = -2 \cdot 1 + 2 \cdot 7 \cdot 3 \cdot 4 = 0 \rightarrow \cos \theta = 0$$

$$25. a. u = (2, 2) \quad v = (4, -1)$$

$$|3 \cdot 4 + 2 \cdot (-1)| \leq \sqrt{3^2 + 2^2} \cdot \sqrt{4^2 + (-1)^2}$$

$$|10| \leq \sqrt{13} \cdot \sqrt{17}$$

✓

$$b. u = (-3, 1, 0), v = (2, -1, 3)$$

$$|-3 \cdot 2 + 1 \cdot (-1) + 0 \cdot 3| \leq \sqrt{3^2 + 1^2 + 0^2} \cdot \sqrt{2^2 + (-1)^2 + 3^2}$$

$$7 \leq \sqrt{10} \cdot \sqrt{14}$$

$$7 \leq \sqrt{140}$$

✓

$$c. u = (0, 2, 2, 1), v = (1, 1, 1, 1)$$

$$|0 \cdot 1 + 2 \cdot 1 + 2 \cdot 1 + 1 \cdot 1| \leq \sqrt{0^2 + 2^2 + 2^2 + 1^2} \cdot \sqrt{1^2 + 1^2 + 1^2 + 1^2}$$

$$5 \leq \sqrt{9} \cdot \sqrt{4}$$

$$5 \leq 6$$

✓

$$27. \|P - P_0\| = 1 \quad P - P_0 = (x - x_0, y - y_0, z - z_0)$$

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = 1^2$$

$$\text{Centro} = P_0 = (x_0, y_0, z_0)$$

$$S = \{ (x, y, z) \in \mathbb{R}^3 \mid (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = 1 \}$$

$$\text{Raio} = r = 1$$

$$29. \|V\| = \sqrt{V_1^2 + V_2^2 + \dots + V_n^2}$$

$$b. \|V\| = 0, \text{ se, e só se, } v_i = 0.$$

a.

$\|V\| \geq 0$, como a quadrado sempre
se é um +, a soma é a soma todos,
então $\|V\| \geq 0$.

De $v_i \geq 0$, a soma sempre mais ou menos
que 0. O único caso é $V = 0$.

$$31.d. (u-v) \cdot w = u \cdot w - v \cdot w$$

$$(u \cdot w - v \cdot w) \rightarrow \text{Distributivität}$$

$$(u-v) \cdot w = \sum_{i=1}^n (u_i - v_i) w_i = \sum_{i=1}^n (u_i w_i - v_i w_i)$$

$$= \sum_{i=1}^n u_i w_i - \sum_{i=1}^n v_i w_i$$

$$32. a(u \cdot v) = u \cdot (a \cdot v)$$

$$a(u_1 v_1 + u_2 v_2 + \dots + u_n v_n)$$

$$a u_1 v_1 + a u_2 v_2 + \dots + a u_n v_n$$

$$a_1(a v_1) + a_2(a v_2) + \dots + a_n(a v_n)$$

$$33. \|u+v\| \leq \|u\| + \|v\|$$

$$(\|u+v\|)^2 = (\|u\| + \|v\|)^2$$

$$\|u\|^2 + 2(u \cdot v) + \|v\|^2 = \|u\|^2 + 2\|u\|\|v\| + \|v\|^2$$

$$2(u \cdot v) = 2\|u\|\|v\|$$

$$u \cdot v = \|u\|\|v\|$$

$$\|u\|\|v\| \cos \theta = \|u\|\|v\|$$

$$\cos \theta = 1$$

$$\theta = 0^\circ$$

Vektoren parallel und in die gleiche Richtung.

Exercício 3.3 - (1, 4, 7, 10, 13, 19, 29, 33, 41) -

1a. $u = (6, 1, 4)$ $(6, 1, 4) \cdot (2, 0, -3)$ Ortogonal.
 $v = (2, 0, -3)$ $12 + 0 - 12$
 0

2a. $u = (0, 0, -1)$ $0 \cdot 1 + 0 \cdot 1 + (-1 \cdot 1)$ Não é ortogonal.
 $v = (1, 1, 1)$ -1

3a. $u = (-6, 0, 4)$ $(-6 \cdot 3) + (0 \cdot 1) + (4 \cdot 6)$ Não é ortogonal.
 $v = (3, 1, 6)$ $-18 + 24$
 6

4a. $u = (2, 4, -8)$ $(2 \cdot 5) + (4 \cdot 3) + (-8 \cdot 7)$ Não é ortogonal.
 $v = (5, 3, 7)$ $-56 + 22$
 -34

4a. $v_1 = (2, 3)$ $(2 \cdot -3) + (3 \cdot 2)$ Conjunto ortogonal.
 $v_2 = (-3, 2)$ 0

5a. $v_1 = (1, -2)$ $(1 \cdot -2) + (-2 \cdot 1)$ Não é um conjunto ortogonal.
 $v_2 = (-2, 1)$ $-2 - 2 = -4$

6a. $v_1 = (1, 0, 1)$ $(1 \cdot 1 \cdot -1) + (0 \cdot 1 \cdot 0) + (1 \cdot 1 \cdot 1)$ É um conjunto ortogonal.
 $v_2 = (1, 1, 1)$ 0
 $v_3 = (-1, 0, 1)$

d. $v_1(2, 2, 1)$

$v_2(2, 1, -2)$

$v_3(1, 2, 2)$

$(2, 2, 1), (-2, 1, 2), (1, -2, 2)$
-9

Não é um conjunto ortogonal
mut.

7. $A=(1, 1, 1)$

$B=(-2, 0, 3)$

$C=(-3, -1, 1)$

$d^2 = \Delta x^2 + \Delta y^2 + \Delta z^2$

$AB: x=2-1=1, y=0-1=-1, z=3-1=2$
 $1^2 + (-1)^2 + (2)^2 = 14$

$y = 0 - 1 = -1$

$z = 3 - 1 = 2$

cateto

$BC: x = -3 + 2 = -1$

$1^2 + (-1)^2 + (-2)^2 = 6$

$y = -1 - 0 = -1$

$z = 1 - 3 = -2$

$AC: x = -3 - 1 = -4$

$(-4)^2 + (-2)^2 + 0^2 = 20$

hipotenusa

$y = -1 - 1 = -2$

$z = 1 - 1 = 0$

Dim. Como a soma do quadrado dos catetos ($14 + 6$) é igual o quadrado da hipotenusa (20).

$a^2 + b^2 = c^2$

$14 + 6 = 20$

10. $P(1, 1, 4); m(1, 9, 8)$

Forma Ponto-normal

$1(x-1) + 9(y-1) + 8(z-4) = 0$

$x-1 + 9y-9 + 8z-32 = 0$

$x + 9y + 8z - 42 = 0$

$$13-4x-y+8z=5 \quad 7x-3y+4z=8$$

$$m=(4,-1,2)$$

$$n=(7,-3,4)$$

$$\text{Razão } x = \frac{4}{7}$$

$$\text{Razão } y = \frac{1}{3}$$

$$\text{Razão } z = \frac{1}{2}$$

$$\frac{4}{7} \neq \frac{1}{3} \neq \frac{1}{2}$$

Não são
paralelas.

$$19. \text{proj}_{\begin{pmatrix} 1, -2 \\ -4, -3 \end{pmatrix}} \begin{pmatrix} 1, -2 \end{pmatrix} = \frac{|u \cdot a|}{\|a\|} \quad \frac{|(1, -2) \cdot (-4, -3)|}{\sqrt{(-4)^2 + (-3)^2}} = \frac{+9}{5} = 0,4.$$

$$29. 4x+3y+4=0 \quad \begin{matrix} -12 & +3 & +4 \\ (4, -3, 1) \end{matrix} \quad \frac{|(4, -3, 1) \cdot (-4, -3, 1)|}{\sqrt{4^2 + (-3)^2}} = \frac{5}{5} = 1 \quad d=1.$$

$$33. x+2y-2z=4 \quad \begin{matrix} 3 & +2 & -2 \\ (3, 1, -2) \end{matrix} \quad \frac{|(3, 1, -2) \cdot (1, 2, -2)|}{\sqrt{3^2 + 2^2 + (-2)^2}} = \frac{5}{3} \quad d = \frac{5}{3}$$

$$41. a. \cos \alpha = \frac{a}{\|v\|} \quad v=(a, b, c) \quad i=(1, 0, 0) = a$$

$$v \cdot i = \|v\| \|i\| \cos \alpha$$

$$\|i\| = 1$$

$$a = \|v\| \cos \alpha$$

$$\cos \alpha = \frac{a}{\|v\|}$$

$$b. \beta = v \cdot j = b \quad b = \|v\| \cdot 1 \cdot \cos \beta \Rightarrow \cos \beta = \frac{b}{\|v\|}$$

$$\gamma = v \cdot k = c \quad c = \|v\| \cdot (1) \cdot \cos \gamma \Rightarrow \cos \gamma = \frac{c}{\|v\|}$$

$$c. \underline{v} = (a, b, c)$$

$$\frac{\underline{v}}{\|\underline{v}\|} = \left(\frac{a}{\|\underline{v}\|}, \frac{b}{\|\underline{v}\|}, \frac{c}{\|\underline{v}\|} \right)$$

$$\frac{\underline{v}}{\|\underline{v}\|} = (\cos \alpha, \cos \beta, \cos \gamma).$$

$$d. (x, y, z) = x^2 + y^2 + z^2$$

$$\|\underline{u}\|^2 = \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma$$

$$\|\underline{u}\|^2 = 1$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Exercício 3.4 (1, 2, 3, 4, 9, 10, 11, 12).

1. $P(-4, 1)$ $V=(0, -8)$

$$P = (-4, 1) + t(0, -8) \quad \begin{cases} x = -4 \\ y = 1 - 8t \end{cases}$$

2. $P(2, -1)$ $V=(-4, -2)$

$$P = (2, -1) + t(-4, -2) \quad \begin{cases} x = 2 - 4t \\ y = -1 - 2t \end{cases}$$

3. $P(0, 0, 0)$ $V=(-3, 0, +1)$

$$P = (0, 0, 0) + t(-3, 0, +1) \quad \begin{cases} x = -3t \\ y = 0 \\ z = +t \end{cases}$$

4. $P(-9, 3, 4)$ $V=(-1, 6, 0)$

$$P = (-9, 3, 4) + t(-1, 6, 0) \quad \begin{cases} x = -9 - t \\ y = 3 + 6t \\ z = 4 \end{cases}$$

9. $P(-3, 1, 0)$ $P = (-3, 1, 0) + s(0, -3, 6) + t(-5, 1, 2)$

$$V_1 = (0, -3, 6)$$

$$V_2 = (-5, 1, 2)$$

$$\begin{cases} x = -3 - 5t \\ y = 1 - 3s + t \\ z = 6s + 2t \end{cases}$$

10. $P(0,6,-2) \quad V_1(0,9,-1) \quad V_2(0,-3,0)$

$$P = (0,6,-2) + s(0,9,-1) + t(0,-3,0)$$

$$\begin{cases} x = 0 \\ y = 6 + 9s - 3t \\ z = -2 - s \end{cases}$$

11. $P(-1,1,4) \quad V_1(6,-1,0) \quad V_2(-1,3,1)$

$$P = (-1,1,4) + s(6,-1,0) + t(-1,3,1)$$

$$\begin{cases} x = -1 + 6s - t \\ y = 1 - s + 3t \\ z = 4 + t \end{cases}$$

12. $P(0,5,-4) \quad V_1(0,0,-5) \quad V_2(1,-3,-2)$

$$P = (0,5,-4) + s(0,0,-5) + t(1,-3,-2)$$

$$\begin{cases} x = t \\ y = 5 - 3t \\ z = -4 - 5s - 2t \end{cases}$$

Questão 3.5 (3, 4, 11, 12, 13, 15, 25, 26, 36) -

3. $u = (-6, 4, 2)$

$v = (3, 1, 5)$

$$u \times v = \begin{vmatrix} i & j & k \\ -6 & 4 & 2 \\ 3 & 1 & 5 \end{vmatrix}$$

$i = (4 \cdot 5) - (2 \cdot 1) = 18$

$j = (-6 \cdot 5) - (2 \cdot 3) = -36$

$k = (-6 \cdot 1) - (4 \cdot 3) = -18$

$w = (18, 36, -18)$

4. $u = (1, 1, -2)$

$v = (2, -1, 2)$

$$u \times v = \begin{vmatrix} i & j & k \\ 1 & 1 & -2 \\ 2 & -1 & 2 \end{vmatrix}$$

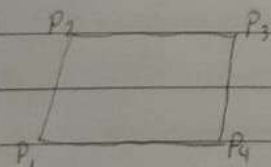
$i = (1 \cdot 2) - (-2 \cdot -1) = 0$

$j = -(1 \cdot 2) - (-2 \cdot 2) = 6$

$k = (1 \cdot -1) - (1 \cdot 2) = -3$

$w = (0, -6, -3)$

11. $P_1(1, 2), P_2(4, 4), P_3(7, 5), P_4(4, 3)$



Adjacências: (P_1, P_2) (P_1, P_4)

$A_{area} = 3$

$u = P_2 - P_1 = (4 - 1, 4 - 2) = (3, 2)$

$v = P_4 - P_1 = (4 - 1, 3 - 2) = (3, 1)$

$A = |\det u, v| = (3 \cdot 1) - (2 \cdot 3) = |-3| = 3$

12. $P_1(3, 2), P_2(5, 4), P_3(9, 4), P_4(7, 2)$



$u = P_2 - P_1 = (5 - 3, 4 - 2) = (2, 2)$

$v = P_4 - P_1 = (7 - 3, 2 - 2) = (4, 0)$

$A = |\det u, v| = (2 \cdot 0) - (0 \cdot 4) = |-2| = 2$

13. $A(2, 0)$

$B(3, 4)$

$C(-1, 2)$

$$D = \begin{vmatrix} 2 & 0 & 1 \\ 3 & 4 & 1 \\ -1 & 2 & 1 \end{vmatrix}$$

$A = \frac{1}{|D|} \Rightarrow \frac{1}{2} \cdot 14 \Rightarrow 7$

$A = 7$

$(8 \cdot 0 + 6) - (-4 \cdot 4 + 0) = 14$

$$15. P_1 = (2, 6, 1) \quad P_2 = (1, 1, 1) \quad P_3 = (4, 6, 2)$$

$$u \times v = \begin{vmatrix} i & j & k \\ -1 & -5 & 2 \\ 2 & 0 & 3 \end{vmatrix}$$

$$u \times P_2 - P_1 = (1-2, 1-6, 1-1) = (-1, -5, 2)$$

$$m. P_3 - P_1 = (4-2, 6-6, 2-1) = (2, 0, 3)$$

$$i = (-5 \cdot 3) - (2 \cdot 0) = -15$$

$$j = -((-1 \cdot 3) - (2 \cdot 2)) = 7$$

$$k = (-1 \cdot 0) - (-5 \cdot 2) = 10$$

$$V = (-15, 7, 10)$$

$$\|u \times v\| = \sqrt{(-15)^2 + 7^2 + 10^2} = \sqrt{225 + 49 + 100} = \sqrt{374}$$

$$A = \frac{\sqrt{374}}{2} \approx 9,67$$

$$25. a. u \cdot (w \times v) = -u \cdot (v \times w) = -3 \quad u \cdot (w \times v) = -u \cdot (v \times w)$$

$$b. (v \times w) \cdot u = 3 \quad (A \cdot B = B \cdot A)$$

$$c. w \cdot (u \times v) = 3 \quad (u, v, w) = (w, u, v)$$

$$26. a. v \cdot (u \times w) = -3 \quad \text{Não cíclica}$$

$$b. (u \times w) \cdot v = -3 \quad \text{Não cíclica comutada}$$

$$c. v \cdot (w \times w) = 0 \quad \text{Produto vet de um vet por ele mesmo é 0.}$$

$$36. A_b = \frac{1}{2} |b \times c| \quad V = \frac{1}{3} \left(\frac{1}{2} |b \times c| \right) \left(\frac{|a \cdot (b \times c)|}{|b \times c|} \right)$$

$$h = a \cdot \cos \theta$$

$$V = \frac{1}{6} |a \cdot (b \times c)|$$

$$h = \frac{|a \cdot (b \times c)|}{|b \times c|}$$